

COMP1511

Introduction to Discrete Mathematics

Lecture Notes

2023/24

The three main topics that are covered in this module are:

1. Combinatorics: Counting Methods
2. Discrete Probability Theory
3. Graph Theory

The lecture notes closely follow the material presented in Chapters 6, 7, 10 and 11 of the accompanying book (that is referred to in the rest of the text as *the book*):

K. H. Rosen, **Discrete Mathematics and its Applications**, 7th Edition, McGraw Hill Education, 2013.

All of the material from the module syllabus will be covered in the lecture notes and presented in the lectures. The examination and coursework will only assess topics covered in the module syllabus. You should use the book as an additional source of problems to solve and as an alternative explanation of a topic. The book is there to supplement the lecture notes. Not all of the material from the chapters is covered in the lecture notes (i.e. lectures), and the lectures sometimes cover the material in a different order than the one in the book.

We expect you, as a student, to:

1. Attend lectures and carefully read the lecture notes.
2. Attend and engage in tutorials, completing all problems requested by your tutorial tutor.
3. Submit all courseworks.
4. To truly absorb the material covered, you should also practice exercises from the relevant sections of the book (the odd numbered ones have solutions in the back).

1 Combinatorics: Counting Methods

(Corresponds to Sections 6.1 - 6.5 in the book)

Combinatorics is the study of arrangements of objects, and it is an important part of discrete mathematics. It was studied as long ago as the seventeenth century, when combinatorial questions arose in the study of gambling games. The counting of objects with certain properties is an important part of combinatorics. It is used, for example, in determining the complexity of algorithms, determining whether there are enough Internet protocol addressees to meet the demand, and when probabilities of events are computed.

1.1 Basic Counting Principles

(Corresponds to Section 6.1 in the book)

The Multiplication Principle for Counting Procedures (or the Product Rule):

If a procedure can be described as a sequence of m independent steps, with i_1 possible outcomes for the first step, i_2 possible outcomes for the second step, ..., and i_m possible outcomes for the m^{th} step, then the total number of composite outcomes of the procedure is $i_1 i_2 \cdots i_m$.

A *sequential counting procedure* is one that satisfies the following three properties:

1. The steps are *ordered*, i.e. any two sequences of different outcomes represents distinguishable composite outcomes.
2. The steps are *independent*, i.e. the number of outcomes of one step is not affected by the outcomes of the preceding steps.
3. The steps are *complete*, i.e. each composite outcome consists of a complete sequence of individual outcomes, one for each step.

Example

A binary sequence is a sequence whose elements come from the set $\{0, 1\}$. For example, 100110 is a binary sequence of length 6.

- (a) How many binary sequences of length 5 are there?
- (b) How many binary sequences of length 5 begin with 101?

Sequential counting procedure for (a):

1. Choose an element for the first position in the sequence. (2 choices)
2. Choose an element for the second position in the sequence. (2 choices)
3. Choose an element for the third position in the sequence. (2 choices)
4. Choose an element for the fourth position in the sequence. (2 choices)
5. Choose an element for the fifth position in the sequence. (2 choices)

⇒ By the Multiplication Principle, the total number of sequences in (a) is $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^5 = 32$.

Another sequential counting procedure for (a):

1. Choose an element for the fifth position in the sequence. (2 choices)
2. Choose an element for the fourth position in the sequence. (2 choices)
3. Choose an element for the third position in the sequence. (2 choices)
4. Choose an element for the second position in the sequence. (2 choices)
5. Choose an element for the first position in the sequence. (2 choices)

⇒ By the Multiplication Principle, the total number of sequences in (a) is $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^5 = 32$.

Sequential counting procedure for (b):

1. Choose the first three elements of the sequence. (1 choice, since they are fixed)
2. Choose an element for the fourth position in the sequence. (2 choices)
3. Choose an element for the fifth position in the sequence. (2 choices)

⇒ By the Multiplication Principle, the total number of sequences in (b) is $1 \cdot 2 \cdot 2 = 2^2 = 4$.

Example

How many ways are there to answer a true-false test with 12 questions.

(a) Every question must be answered.

Sequential counting procedure:

1. Answer first question. (2 choices)
2. Answer second question. (2 choices)
- ...
12. Answer twelfth question. (2 choices)

⇒ By the Multiplication Principle, the total number of ways to answer is $2 \cdot 2 \cdots 2 \cdot 2 = 2^{12}$.

(b) Any number of questions may be left unanswered.

3 choices for each question (true, false, unanswered)

$\Rightarrow 3^{12}$ ways to answer the test.

Example

A licence plate consists of a sequence of 3 letters followed by 3 digits. How many licence plates are possible?

Sequential counting procedure:

1. Choose the first letter. (26 choices)
2. Choose the second letter. (26 choices)
3. Choose the third letter. (26 choices)
4. Choose the first digit. (10 choices)
5. Choose the second digit. (10 choices)
6. Choose the third digit. (10 choices)

$\Rightarrow 26^3 \cdot 10^3 = 17,576,000$ possibilities

Example

(a) How many 7 letter strings are there?

26^7

(b) How many 7 letter palindromes are there? A palindrome is a string that reads the same forward and backward.

Sequential counting procedure:

1. Choose the first letter. (26 choices)
2. Choose the second letter. (26 choices)
3. Choose the third letter. (26 choices)
4. Choose the fourth letter. (26 choices)
5. Choose the remaining three letters. (1 choice, since they are already determined by the choice of the first three letters)

$\Rightarrow$ There are 26^4 palindromes (by the Multiplication Principle).

Note that these steps are independent because the **number** of choices in step 5 is independent of the choices made in previous steps, although the actual choice made in step 5 does depend on the choices made in steps 1-3.

Example (k -element sequences)

Let $a_1, a_2, \dots, a_k$ be a k -element sequence or a k -tuple (note that the order of the elements is important). Sometimes k -tuples are also denoted by $(a_1, a_2, \dots, a_k)$. If there are n choices for each element of the sequence, then by the Multiplication Principle, the number of different k -element sequences is n^k .

Example

Find the number of distinct divisors of the integer 64800.

$64800 = 2^5 3^4 5^2$ prime factorization

Any divisor of 64800 must have a prime factorization of the form $2^i 3^j 5^k$ where $0 \leq i \leq 5, 0 \leq j \leq 4, 0 \leq k \leq 2$.

Sequential counting procedure:

1. Choose i . (6 choices)
2. Choose j . (5 choices)
3. Choose k . (3 choices)

$\Rightarrow$ Total number of different divisors is $6 \cdot 5 \cdot 3 = 90$.

Example

6 different books are arranged on a bookshelf.

(a) How many distinguishable orders of arrangements are there?

Consider the following counting procedure:

1. Choose one book for the first position. (6 choices)
2. Choose a different book for the second position. (5 choices)
- ...
6. Choose the last book for the sixth position. (1 choice)

$\Rightarrow$ Total number of outcomes is $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$

(Note that these steps are considered to be independent in the sense that the number of possible outcomes is always the same, regardless of the previous outcomes, even though the actual choices do depend on the previous outcomes.)

(b) Suppose that 2 of the 6 books are blue and should not be placed next to each other. How many different arrangements are there with 2 blue books not adjacent?

Note: the above steps are not independent. But we can count the arrangements that we **do not want** as follows:

1. Choose 2 consecutive positions for the blue books. (5 choices)
 2. Arrange blue books in these positions. (2 ways)
 3. Arrange the 4 remaining books in remaining positions. ($4 \cdot 3 \cdot 2 \cdot 1 = 24$ ways)
- $\Rightarrow$ Number of arrangements we do not want is $5 \cdot 2 \cdot 24 = 240$.

$\Rightarrow$ Number of arrangements we want is $720 - 240 = 480$.

Example

Consider the problem of determining the number of outcomes when n dice are rolled.

(a) How many outcomes are possible when n dice are rolled if each die has a different color or is somehow different from the other?

There are 6 possible outcomes for each die.

Each composite outcome can be listed as an n -tuple, with each n -tuple representing a distinguishable outcome.

$\Rightarrow$ There are 6^n different outcomes.

(b) How many outcomes when 2 identical dice are rolled?

There are $6^2 = 36$ different ordered pairs whose coordinates are integers from 1 to 6, but not all of them represent distinguishable outcomes.

An outcome when one die is i the other j , $i \neq j$, is represented by two different ordered pairs: (i, j) and (j, i) .

There are 6 outcomes that are represented by a single ordered pair ((i, i) for $i = 1, \dots, 6$) and 15 outcomes that are represented by 2 ordered pairs.

$\Rightarrow$ There are 21 different outcomes.

Example

What is the value of k after the following code, where $n_1, \dots, n_m$ are positive integers, has been executed?

```
k := 0
for i1 := 1 to n1
    for i2 := 1 to n2
```

```

      .
      .
      .
  for  $i_m := 1$  to  $n_m$ 
     $k := k + 1$ 

```

The initial value of k is 0. Each time the nested loop is executed, 1 is added to k . Let T_j be the task of traversing the j^{th} loop. Then the number of times the nested loop is traversed is the number of ways to do tasks $T_1, \dots, T_m$. The number of ways to carry out task T_j , $j = 1, \dots, m$, is n_j because the j^{th} loop is traversed once for each integer i_j with $1 \leq i_j \leq n_j$.

$\Rightarrow$ By the Multiplication Principle, the nested loop is traversed $n_1 n_2 \cdots n_m$ times.

$\Rightarrow$ After the code is executed $k = n_1 n_2 \cdots n_m$.

The Addition Principle for Counting Procedures (or the Sum Rule): If a procedure can be broken up into m events or cases whose sets of outcomes are mutually exclusive, with j_1 possible outcomes for the first event, j_2 possible outcomes for the second event, $\dots$, and j_m possible outcomes for the m^{th} event, then the total number of outcomes of the procedure is $j_1 + j_2 + \dots + j_m$.

Example

On a bookshelf there are 5 different calculus books, 6 different linear algebra books and 7 different combinatorics books.

(a) In how many ways can a pair of books of different types be chosen?

3 mutually exclusive cases can occur:

Calculus - linear algebra pair is chosen. ($5 \cdot 6$ choices)

Linear algebra - combinatorics pair is chosen. ($6 \cdot 7$ choices)

Calculus - combinatorics pair is chosen. ($5 \cdot 7$ choices)

$\Rightarrow$ There are $5 \cdot 6 + 6 \cdot 7 + 5 \cdot 7 = 107$ different outcomes.

(Note: here we used both the Multiplication and the Addition principle.)

(b) In how many ways can a single book of each type be chosen?

Count as follows:

1. Choose a calculus book. (5 choices)

2. Choose a linear algebra book. (6 choices)

3. Choose a combinatorics book. (7 choices)

$\Rightarrow$ There are $5 \cdot 6 \cdot 7 = 210$ different outcomes.

(Note: here we used just the Multiplication Principle.)

Example

How many integers are there from 1 through 9999 that have distinct digits?

Divide integers into 4 types: 1-digit integers, 2-digit integers, 3-digit integers and 4-digit integers.

Choose individual digits starting from left to right. Each digit is chosen from $\{0, 1, \dots, 9\}$, except that the first digit cannot be 0 and each succeeding digit must be different from previously chosen one.

• 4-digit integers

1. Choose the first digit. (9 choices, since it cannot be 0)

2. Choose the second digit. (9 choices, since it cannot equal the first digit)

3. Choose the third digit. (8 choices, since it cannot equal the first two digits)

4. Choose the fourth digit. (7 choices, since it cannot equal the first three digits)

$\Rightarrow$ There are $9 \cdot 9 \cdot 8 \cdot 7$ 4-digit integers that have distinct digits.

• 3-digit integers

1. Choose the first digit. (9 choices, since it cannot be 0)

2. Choose the second digit. (9 choices, since it cannot equal the first digit)

3. Choose the third digit. (8 choices, since it cannot equal the first two digits)

$\Rightarrow$ There are $9 \cdot 9 \cdot 8$ 3-digit integers that have distinct digits.

• 2-digit integers

1. Choose the first digit. (9 choices, since it cannot be 0)

2. Choose the second digit. (9 choices, since it cannot equal the first digit)

$\Rightarrow$ There are $9 \cdot 9$ 2-digit integers that have distinct digits.

• 1-digit integers

1. Choose the first digit. (9 choices, since it cannot be 0)

$\Rightarrow$ There are 9 1-digit integers that have distinct digits.

$\Rightarrow$ Total = $9 \cdot 9 \cdot 8 \cdot 7 + 9 \cdot 9 \cdot 8 + 9 \cdot 9 + 9$

Example

What is the value of k after the following code, where $n_1, \dots, n_m$ are positive integers, has been executed?

```

 $k := 0$ 
for  $i_1 := 1$  to  $n_1$ 
     $k := k + 1$ 
for  $i_2 := 1$  to  $n_2$ 
     $k := k + 1$ 
    .
    .
    .
for  $i_m := 1$  to  $n_m$ 
     $k := k + 1$ 

```

The initial value of k is 0. The code is made up of m independent loops (traversed one at a time). Each time a loop is executed, 1 is added to k . There are n_i ways to traverse the i^{th} loop, for $i = 1, \dots, m$.

$\Rightarrow$ By the Addition Principle, the number of ways to traverse the m loops is $n_1 + n_2 + \dots + n_m$.

$\Rightarrow$ After the code is executed $k = n_1 + n_2 + \dots + n_m$.

Counting What You Do Not Want**Example**

How many binary sequences of length 5 have at least two 1's?

The total number of binary sequences of length 5 is 2^5 .

There is one sequence with no 1's and 5 sequences with exactly one 1. So the number of sequences that we do not want is $1 + 5 = 6$.

$\Rightarrow$ The number of sequences that we do want is $2^5 - 6$.

Example

In a standard deck of cards there are 4 suits: clubs, diamonds, hearts and spades. Each suit contains 13 cards: 2, 3, $\dots$, 10, J, Q, K, A. So there are $4 \cdot 13 = 52$ cards in the deck.

Suppose we choose a sequence of 4 cards from the deck, with replacement, i.e. one card at a time is chosen and it is placed back into the deck before the next

choice.

(a) In how many ways can a sequence of 4 cards be chosen, with replacement, so that some card is repeated?

Number of ways to choose a sequence with replacement = 52^4

Number of sequences with no repetition = $52 \cdot 51 \cdot 50 \cdot 49$

Number of ways to choose a sequence with some card repeated = $52^4 - 52 \cdot 51 \cdot 50 \cdot 49$

(b) In how many ways can a sequence of 4 cards be chosen, with replacement, so that the fourth card is the first repeated card?

1. Choose the first card. (52 choices)
 2. Choose the second card. (51 choices, since it must be different from the first card)
 3. Choose the third card. (50 choices, since it must be different from the first two cards)
 4. Choose the fourth card. (3 choices, since it must be equal to one of the first three cards)
- $\Rightarrow$ There are $52 \cdot 51 \cdot 50 \cdot 3$ such sequences.

(c) In how many ways can a sequence of 4 cards be chosen, with replacement, so that the fourth card is a repeat, but not necessarily the first repeat?

Count what you do not want:

To choose a sequence where the fourth card is not a repeat, first choose the fourth card, and then choose the first three cards to be any cards different from the fourth.

$\Rightarrow$ What we do not want = $52 \cdot 51^3$.

What we want = $52^4 - 52 \cdot 51^3$.

Principle of Inclusion-Exclusion: Let A_1 and A_2 be two sets. The number of ways to select an element from A_1 is $|A_1|$, the number of ways to select an element from A_2 is $|A_2|$, and the number of ways to select an element from A_1 or A_2 is $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$.

Example

How many binary sequences of length 6 start with 1 **or** end with 00?

Note that the events “a binary sequence of length 6 starts with 1” and “a binary sequence of length 6 ends with 00” do not have mutually exclusive outcomes and so we cannot apply the Addition Principle. Instead, we use the Principle of Inclusion-Exclusion.

Let A_1 be the set of binary sequences of length 6 that start with 1.

Let A_2 be the set of binary sequences of length 6 that end with 00.

We want $|A_1 \cup A_2|$.

$|A_1| = 2^5$ (since sequences in A_1 must start with 1), $|A_2| = 2^4$ (since sequences in A_2 must end with 00) and $|A_1 \cap A_2| = 2^3$ (since sequences in $A_1 \cap A_2$ must start with 1 and end with 00).

$$\Rightarrow |A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2| = 2^5 + 2^4 - 2^3$$

1.2 Selections

(Corresponds to Sections 6.3 and 6.5 in the book)

Suppose that a child is offered a bag containing three types of sweets, aniseed drops (A), butter mints (B) and cherry drops (C). In how many ways can the child select two sweets?

This problem is not defined well, since we are not told whether or not the same type of sweet can be selected twice (i.e. is AA allowed?), and does it matter in which order the sweets are selected (i.e. is AB to be treated as a different selection from BA, or not?).

There are 4 cases to consider:

1. Repeats allowed and order matters. This gives 9 possibilities:
AA, AB, AC, BA, BB, BC, CA, CB, CC.
2. Repeats allowed and order does not matter. This gives 6 possibilities:
AA, AB, AC, BB, BC, CC.
3. Repeats not allowed and order matters. This gives 6 possibilities:
AB, AC, BA, BC, CA, CB.

4. Repeats not allowed and order does not matter. This gives 3 possibilities:
AB, AC, BC.

1.2.1 Permutations

Consider the selection of k objects from a set of n objects where **order matters and repetition is not allowed**. Any such selection is called a k -permutation. The total number of k -permutations of a set of n (distinct) objects is denoted by $P(n, k)$.

Example

$$X = \{a, b, c, d\}$$

2-permutations are: $ab, ac, ad, ba, bc, bd, ca, cb, cd, da, db, dc$

Factorial notation: for a non-negative integer n , n factorial is defined as

$$n! = \begin{cases} 1 & \text{if } n = 0 \\ n(n-1)(n-2) \dots 1 & \text{if } n \geq 1. \end{cases}$$

Theorem 1.1 For integers k and n such that $0 \leq k \leq n$, the number of k -permutations of a set of n objects is $P(n, k) = \frac{n!}{(n-k)!}$.

Proof: First suppose that $k = 0$. Then $P(n, 0) = 1$ and $\frac{n!}{(n-0)!} = \frac{n!}{n!} = 1$, and hence the result holds. Now assume that $1 \leq k$, and consider the following counting procedure:

1. Choose an element for the first position. (n choices)
2. Choose an element for the second position. ($n - 1$ choices)
- ...
- k . Choose an element for the k^{th} position. ($n - k + 1$ choices)

So by the Multiplication Principle, $P(n, k) = n(n-1) \dots (n-k+1)$.

$$\frac{n!}{(n-k)!} = \frac{n(n-1) \dots (n-k+1)(n-k)!}{(n-k)!} = n(n-1) \dots (n-k+1), \text{ and hence the result holds.}$$

□

Note that $P(n, n) = n!$.

Example

How many 7 letter strings with no repeated letters can be formed from English alphabet?

7 letter strings are 7-permutations of 26 letters, since the order is important and no repeated letters are allowed. So we want $P(26, 7)$.

By Theorem 1.1, $P(26, 7) = 26 \cdot 25 \cdot 24 \cdot 23 \cdot 22 \cdot 21 \cdot 20$

Example

In how many ways can 5 books be chosen from a pile of 10 distinct books and arranged on a bookshelf?

The arrangements are 5-permutations of 10 distinct books, since the order on the bookshelf is important and repetition is not possible.

$\Rightarrow$ The number of arrangements is $P(10, 5) = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6$.

Example

A computer center has to run 9 different programs in a row, 4 of them in FORTRAN and 5 in pascal. Find the number of possible schedules for running the programs if fortran programs must be first, followed by pascal programs.

1. Arrange fortran programs in first 4 positions. ($4!$ ways)

2. Arrange pascal programs in last 5 positions. ($5!$ ways)

$\Rightarrow$ There are $4!5!$ schedules.

1.2.2 Combinations

Consider the selection of k objects from a set of n objects where **order does not matter and repetition is not allowed**. Any such selection is called a k -combination. The total number of k -combinations of a set of n (distinct) objects is denoted by $C(n, k)$ or $\binom{n}{k}$ (“ n choose k ”).

Example

$X = \{a, b, c, d\}$

2-combinations are: $\{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}$

Theorem 1.2 For integers k and n such that $0 \leq k \leq n$, the total number of k -combinations of a set of n objects is $C(n, k) = \frac{n!}{k!(n-k)!}$.

Proof: Since a set of k elements can be arranged into exactly $k!$ sequences, $P(n, k) = C(n, k)k!$

i.e. count k -permutations by the following procedure:

1. Choose a subset of k different objects from a set with n elements. ($C(n, k)$ ways)

2. Arrange objects sequentially. ($k!$ ways)

$$\Rightarrow P(n, k) = C(n, k)k!$$

$$\Rightarrow C(n, k) = \frac{P(n, k)}{k!} = \frac{n!}{(n-k)!k!} \text{ (by Theorem 1.1).} \quad \square$$

Note that by Theorem 1.2, $C(n, k) = C(n, n - k)$.

Example

(a) How many subsets of three integers can be chosen from the set $\{1, 2, \dots, 10\}$?

Since order does not matter and repetition is not allowed, we want $C(10, 3)$.

$$\text{By Theorem 1.2, } C(10, 3) = \frac{10 \cdot 9 \cdot 8}{3!} = 120$$

(b) How many subsets in (a) contain 3?

Construct subsets by the following procedure:

1. Choose 3. (1 way)

2. Choose any two of the remaining integers. ($C(9, 2)$ ways)

$$\Rightarrow 1 \cdot C(9, 2) = 36$$

(c) How many of the subsets in (a) contain at least one multiple of 3?

Count what we do not want first:

The multiples of 3 in $\{1, \dots, 10\}$ are 3, 6 and 9, so the number of subsets that do not contain any multiple of 3 is $C(7, 3) = 35$.

$$\Rightarrow \text{What we want} = 120 - 35 = 85.$$

Example (counting subsets of a finite set)

Number of ways to choose a subset from a set X of n objects:

(a) Any subset: to choose a subset, for each element of X decide whether it is in the subset or not (so 2 choices for each element of X).

$$\Rightarrow 2^n \text{ ways}$$

(b) Any nonempty subset: $2^n - 1$ ways

(c) Any k -element subset: $C(n, k)$ ways

Example

A new club starts with n different members. If each member shakes hands with everyone else exactly once, how many handshakes take place?

Number of handshakes

= number of unordered pairs of different members

$$= C(n, 2) = \frac{n(n-1)}{2}.$$

Example

Suppose a juggling club has 17 members.

(a) In how many ways can they choose 4 different members to be president, vice president, secretary and treasurer?

$$P(17, 4) = \frac{17!}{13!}$$

(b) In how many ways can they choose 4 different members to form a juggling team?

The 4 positions on the team are identical, so $C(17, 4) = \frac{17!}{4!13!}$.

(c) In how many ways can a captain and 3 regular members be chosen?

1. Choose a captain. (17 ways)

2. Choose the remaining 3 members. ($C(16, 3)$ ways)

$\Rightarrow$ There are $17 \cdot C(16, 3) = 17 \cdot \frac{16!}{3!13!}$ choices.

Example

How many 4-digit integers are there where each digit is greater than those to the right?

1. Choose 4 distinct digits. ($C(10, 4)$ ways)

2. Arrange them in decreasing order from left to right. (1 way)

$\Rightarrow$ There are $C(10, 4) \cdot 1 = \frac{10!}{4!6!}$ choices.

1.2.3 Permutations with Unlimited Repetition

Consider the selection of k objects from a set of n objects where **order matters and repetition is allowed**. Any such selection is called a *k-permutation with unlimited repetition* of a set of n objects.

Theorem 1.3 For positive integers k and n , the number of k -permutations with unlimited repetition of a set of n objects is n^k .

Proof: Consider the following counting procedure:

1. Choose the first element of the sequence. (n choices)
 2. Choose the second element of the sequence. (n choices)
 - ...
 - k . Choose the k^{th} element of the sequence. (n choices)
- $\Rightarrow$ So by the Multiplication Principle, there are n^k choices. □

Example

How many 5 letter strings can be formed from English alphabet?

Since order matters and repetition is allowed, by Theorem 1.3, the number of 5 letter strings is 26^5 .

Example

Let $A = \{1, \dots, 6\}$ and $B = \{a, b, c, d, e\}$.

(a) Find the number of functions from A to B .

1. Choose an element that 1 maps to. (5 choices)
 2. Choose an element that 2 maps to. (5 choices)
 - ...
 6. Choose an element that 6 maps to. (5 choices)
- $\Rightarrow$ There are 5^6 functions.

(b) Find the number of one-to-one functions from B to A .

1. Choose an element that a maps to. (6 choices)
2. Choose an element that b maps to. (5 choices)
- ...

5. Choose an element that e maps to. (2 choices)

$\Rightarrow$ There are $P(6, 5) = \frac{6!}{1!} = 6!$ such functions.

(c) Find the number of onto functions from A to B .

1. Select 2-element subset of A . ($C(6, 2)$ ways)

2. Assign images without repetition to the 2-element subset and 4 remaining individual elements of A . ($5!$ ways)

$\Rightarrow$ There are $C(6, 2) \cdot 5! = \frac{6!}{2!4!} \cdot 5!$ such functions.

1.2.4 Combinations with Unlimited Repetition

Consider the selection of k objects from a set of n objects where **order does not matter and repetition is allowed**. Any such selection is called a *k-combination with unlimited repetition* of a set of n objects.

We will show the following:

The number of k -combinations with unlimited repetition of a set of n objects

= the number of nonnegative integer solutions of $x_1 + x_2 + \dots + x_n = k$

= the number of distributions of k identical balls into n distinct boxes

= the number of sequences of length $k + n - 1$ consisting of k 0's and $n - 1$ |'s.

Let A be the set of all k -combinations with unlimited repetition of a set X with n elements.

Let B be the set of all nonnegative integer solutions of $x_1 + x_2 + \dots + x_n = k$.

Let C be the set of all distributions of k identical balls into n distinct boxes.

Let D be the set of all sequences of length $k + n - 1$ consisting of k 0's and $n - 1$ |'s.

So we want to show that $|A| = |B| = |C| = |D|$.

• We first show that $|A| = |B|$.

Let $X = \{1, 2, \dots, n\}$. Suppose that we select x_1 objects of type 1, x_2 objects of type 2, $\dots$ and x_n objects of type n . Then, for an n -tuple of integers $(x_1, x_2, \dots, x_n)$ the following holds:

for every $i \in \{1, \dots, n\}$, $x_i \geq 0$, and $x_1 + x_2 + \dots + x_n = k$.

Each n -tuple of nonnegative integers that is a solution of the linear equation

$x_1 + x_2 + \dots + x_n = k$ corresponds to exactly one k -combination with unlimited repetition and vice versa.

So there is a one-to-one correspondence between A and B .

$$\Rightarrow |A| = |B|$$

• Next we show that $|B| = |C|$.

Consider a distribution of k identical balls into n distinct boxes, where box 1 gets x_1 balls, box 2 gets x_2 balls, $\dots$ and box n gets x_n balls. Then, for an n -tuple of integers $(x_1, x_2, \dots, x_n)$ the following holds:

for every $i \in \{1, \dots, n\}$, $x_i \geq 0$, and $x_1 + x_2 + \dots + x_n = k$.

Each n -tuple of nonnegative integers that is a solution of the linear equation $x_1 + x_2 + \dots + x_n = k$ corresponds to exactly one distribution of k identical balls into n distinct boxes, and vice versa.

So there is a one-to-one correspondence between B and C .

$$\Rightarrow |B| = |C|$$

Example

In how many ways can 7 identical balls be distributed into 3 distinct boxes?

00		0		0000
box 1		box 2		box 3

This corresponds to: 2 balls in box 1, 1 ball in box 2 and 4 balls in box 3.

A sequence of 7 0's and 2 |'s can be thought of as a distribution of 7 identical balls into 3 distinct boxes.

3 balls in box 1

0 balls in box 2

4 balls in box 3

This can be modeled with the sequence: 000||0000.

Every distribution of 7 identical balls into 3 distinct boxes can be modeled with a sequence of 7 0's and 2 |'s.

So there is a one-to-one correspondence between the distributions of 7 identical balls into 3 distinct boxes and the sequences of 7 0's and 2 |'s.

$\Rightarrow$ The number of distributions of 7 identical balls into 3 distinct boxes = the number of sequences of 7 0's and 2 |'s.

Consider the following procedure for counting the number of sequences of 7 0's and 2 |'s:

1. Choose 7 positions in the sequence for the 0's. ($C(9, 7)$ ways)
2. Choose the remaining 2 positions in the sequence for the |'s. (1 way)

$\Rightarrow$ There are $C(9, 7) = \frac{9!}{7!2!}$ such sequences.

• In general, $|C| = |D|$.

Theorem 1.4 For positive integers k and n , the number of k -combinations with unlimited repetition of a set of n objects is $C(k + n - 1, k) = \frac{(k+n-1)!}{k!(n-1)!}$.

Proof: The number of k -combinations with unlimited repetition chosen from a set of n objects is the same as the number of sequences of length $k + n - 1$ consisting of k 0's and $n - 1$ |'s.

Consider the following way of counting such sequences: out of $k + n - 1$ positions, choose k to put 0's in (then put the |'s in the remaining positions), or alternatively, out of $k + n - 1$ positions choose $n - 1$ to put |'s in (then put the 0's in the remaining positions).

$\Rightarrow$ The number of such sequences is $C(k + n - 1, k) = C(k + n - 1, n - 1)$.

By Theorem 1.2, $C(k + n - 1, k) = \frac{(k+n-1)!}{k!(n-1)!}$. □

Example

Find the number of nonnegative integer solutions of $x_1 + x_2 + \dots + x_5 = 30$.

Same as the number of distributions of 30 identical balls into 5 distinct boxes.

Same as the number of sequences of length $30 + 5 - 1$ consisting of 30 0's and $5 - 1$ |'s, which is

$$C(30 + 5 - 1, 30) = C(34, 30) = \frac{34!}{30!4!}$$

Example

Find the number of positive integer solutions of $x_1 + x_2 + \dots + x_5 = 30$.

Same as distributing 30 identical balls into 5 distinct boxes so that no box is left

empty.

First put one ball in each of the boxes, and then distribute the remaining 25 balls.

$$C(25 + 5 - 1, 25) = C(29, 25) = \frac{29!}{25!4!}$$

Example

A bookshelf holds 11 different books in a row. In how many ways can we choose 4 books so that no 2 consecutive books are chosen?

Represent books by a sequence of 0's and |'s, where | represents a chosen book and 0 represents a book that is not chosen.

0|0|00|000| books 2, 4, 7, 11 are chosen

Want a sequence of 4 |'s and 7 0's but cannot have two consecutive |'s.

The 4 |'s divide the 0's into 5 subsequences. The first and last subsequences may be empty, but second, third and fourth subsequences must contain at least one 0 (because no consecutive books may be chosen).

Same as the number of integer solutions to $x_1 + x_2 + \dots + x_5 = 7$ so that $x_1, x_5 \geq 0$ and $x_2, x_3, x_4 \geq 1$.

Same as the number of ways to distribute 7 identical balls into 5 distinct boxes so that boxes 2, 3 and 4 get at least one ball.

First put one ball in each of the boxes 2, 3 and 4.

	0		0		0	
box 1	box 2	box 3	box 4	box 5		

Then distribute the remaining balls among the 5 boxes.

$$\Rightarrow C(4 + 4, 4) = \frac{8!}{4!4!} \text{ ways}$$

	Ordered selections	Unordered selections
No repetition	$P(n, k)$	$C(n, k)$
Unlimited repetition allowed	n^k	$C(k + n - 1, k)$

1.3 Binomial Coefficients and Combinatorial Identities (Corresponds to Section 6.4 in the book)

At first glance the expression $(a + b)^n$ does not have much to do with combinations, but it turns out that we can obtain the formula for the expansion of $(a + b)^n$ by using the formula for k -combinations of n objects.

$$(a + b)^2 = (a + b)(a + b) = aa + ab + ba + bb = a^2 + 2ab + b^2$$

$$(a + b)^3 = (a + b)(a + b)(a + b) = aaa + aab + aba + abb + baa + bab + bba + bbb = a^3 + 3a^2b + 3ab^2 + b^3$$

Theorem 1.5 Binomial Theorem

If a and b are real numbers and n is a positive integer, then $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$.

Proof: $(a + b)^n = (a + b) \cdots (a + b)$ (n factors)

In the expansion of $(a + b)^n$, a term of the form $a^{n-k} b^k$ arises from choosing b from k factors and a from the other $n - k$ factors. This can be done in $\binom{n}{k}$ ways.

Thus $a^{n-k} b^k$ appears $\binom{n}{k}$ times in the expansion. It follows that

$$\begin{aligned} (a + b)^n &= \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \binom{n}{2} a^{n-2} b^2 + \cdots + \binom{n}{n-1} a^1 b^{n-1} + \binom{n}{n} a^0 b^n \\ &= \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k. \end{aligned} \quad \square$$

The numbers $\binom{n}{k}$ are known as *binomial coefficients* because they appear in the expansion of $(a + b)^n$.

Example

What is the coefficient of $x^{12}y^{13}$ in the expansion of $(x + y)^{25}$?

$$\binom{25}{13} = \frac{25!}{13!12!}$$

Example

Expand $(3x - 2y)^4$ using the Binomial Theorem.

Let $a = 3x$, $b = -2y$ and $n = 4$, and apply the Binomial Theorem.

$$\begin{aligned} (3x - 2y)^4 &= \binom{4}{0} (3x)^4 (-2y)^0 + \binom{4}{1} (3x)^3 (-2y)^1 + \binom{4}{2} (3x)^2 (-2y)^2 + \binom{4}{3} (3x)^1 (-2y)^3 + \binom{4}{4} (3x)^0 (-2y)^4 \\ &= 3^4 x^4 + 4 \cdot 3^3 x^3 (-2y) + 6 \cdot 3^2 x^2 (-2)^2 y^2 + 4(3x)(-2)^3 y^3 + (-2)^4 y^4 \\ &= 81x^4 - 216x^3y + 216x^2y^2 - 96xy^3 + 16y^4 \end{aligned}$$

Example

Find the coefficient of a^4b^5 in the expansion of $(2a - b)^9$.

The term involving a^4b^5 arises in the Binomial Theorem by taking $n = 9$ and $k = 5$: $\binom{9}{5}(2a)^4(-b)^5 = 126 \cdot 2^4 \cdot a^4 \cdot (-1)^5 \cdot b^5 = -2016a^4b^5$

Thus the coefficient of a^4b^5 is -2016.

Example

Find the coefficient of $x^2y^3z^4$ in the expansion of $(x + y + z)^9$.

$(x + y + z)^9 = (x + y + z) \dots (x + y + z)$ (9 factors)

We obtain $x^2y^3z^4$ each time we multiply together x chosen from 2 of the 9 factors, y chosen from 3 of the 9 factors, and z chosen from 4 of the 9 factors.

This can be done in $\binom{9}{2}\binom{7}{3}\binom{4}{4} = \frac{9!}{2!3!4!} = 1260$ ways.

A **combinatorial proof** of an identity is a proof that uses counting arguments to prove that both sides of the identity count the same objects but in different ways or a proof that is based on showing that there is a bijection between the sets of objects counted by the two sides of the identity.

Example

Prove that for a positive integer n , $\sum_{k=0}^n \binom{n}{k} = 2^n$.

Proof using Binomial Theorem:

$$2^n = (1 + 1)^n = \sum_{k=0}^n \binom{n}{k} 1^{n-k} 1^k = \sum_{k=0}^n \binom{n}{k}$$

Combinatorial proof:

A set with n elements has 2^n different subsets.

Each subset has k elements for $k \in \{0, 1, \dots, n\}$.

There are $\binom{n}{k}$ subsets with k elements.

Therefore the total number of subsets is $\sum_{k=0}^n \binom{n}{k}$, and hence $\sum_{k=0}^n \binom{n}{k} = 2^n$.

Theorem 1.6 Pascal's Identity

Let n and k be positive integers with $n \geq k$. Then $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$.

Proof: Let T be a set containing $n + 1$ elements. Let $a \in T$ and $S = T \setminus \{a\}$.

There are $\binom{n+1}{k}$ subsets of T containing k elements.

A subset of T with k elements either contains a together with $k - 1$ elements of S , or contains k elements of S and does not contain a .

So there are $\binom{n}{k-1}$ subsets of T with k elements that contain a , and $\binom{n}{k}$ subsets of T with k elements that do not contain a .

Therefore, $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$. $\square$

Remark: It is also possible to prove Pascal's Identity by algebraic manipulation from the formula for $\binom{n}{k}$.

Theorem 1.7 Vandermonde's Identity

Let m, n and r be nonnegative integers with r not exceeding either m or n . Then

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{r-k} \binom{n}{k}.$$

Proof: Suppose there are m items in one set and n (different) items in a second set.

Then the total number of ways to pick r elements from the union of these sets is $\binom{m+n}{r}$.

Another way to pick r elements from the union is to pick k elements from the first set and then $r - k$ elements from the second set, where k is an integer with $0 \leq k \leq r$.

This can be done in $\binom{n}{k} \binom{m}{r-k}$ ways, using the product rule, and so the total is $\sum_{k=0}^r \binom{m}{r-k} \binom{n}{k}$.

Therefore, $\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{r-k} \binom{n}{k}$. $\square$

1.4 The Pigeonhole Principle (Corresponds to Section 6.2 in the book)

The *Pigeonhole Principle* (also known as the *Dirichlet Principle*) is sometimes useful in answering the question: Is there an item having a given property?

Pigeonhole Principle: If n pigeons fly into k pigeonholes and $k < n$, then some pigeonhole contains at least two pigeons.

Example

Ten persons have first name Alice, Bernard and Charles, and last names Lee, McDuff and Ng. Show that at least two persons have the same first and last names.

There are $3 \cdot 3 = 9$ possible names for the 10 persons. If we think of persons as pigeons and names as pigeonholes, we can consider the assignments of names to people as assigning pigeonholes to pigeons. By the Pigeonhole Principle, some name (pigeonhole) is assigned to at least two persons (pigeons).

Generalized Pigeonhole Principle: If n pigeons fly into k pigeonholes, then some pigeonhole contains at least $\lceil \frac{n}{k} \rceil$ pigeons.

Proof: Suppose that none of the pigeonholes contains more than $\lceil \frac{n}{k} \rceil - 1$ pigeons. Then the total number of pigeons is at most $k(\lceil \frac{n}{k} \rceil - 1) < k((\frac{n}{k} + 1) - 1) = n$, a contradiction. $\square$

Example

Among 100 people there are at least $\lceil \frac{100}{12} \rceil = 9$ who were born in the same month.

Example

What is the minimum number of students required in a discrete mathematics class to be sure that at least 6 will receive the same grade, if there are 5 possible grades, A,B,C,D and F?

The minimum number of students needed to ensure that at least 6 students receive the same grade is the smallest integer n such that $\lceil \frac{n}{5} \rceil = 6$.

The smallest such integer is $n = 5 \cdot 5 + 1 = 26$.

Note that if you have 25 students, it is possible to have 5 who have received each grade, so that no 6 students have received the same grade.

Thus 26 is the minimum number of students needed to ensure that at least 6 students will receive the same grade.

2 Discrete Probability Theory

(Corresponds to Sections 7.1 - 7.3 in the book)

An *experiment* is a process that yields an outcome.

The set S of all possible outcomes of a given experiment is called the *sample space*.

An *event* A is a set of outcomes, i.e. a subset of the sample space S .

Example

Experiment: rolling a six-sided die

Sample space: $S = \{1, 2, 3, 4, 5, 6\}$

Event: obtaining a 4 when rolling a die, i.e. $A = \{4\}$

Since an event is a set, we can combine events to form new events using the various set operations:

- (i) $A \cup B$ is the event that occurs if and only if A occurs *or* B occurs (or both).
- (ii) $A \cap B$ is the event that occurs if and only if A occurs *and* B occurs.
- (iii) $\bar{A}$, the complement of A , is the event that occurs if and only if A does *not* occur.

Two events A and B are called *mutually exclusive* if they are disjoint, i.e. $A \cap B = \emptyset$ (they cannot occur simultaneously). Three or more events are mutually exclusive if every two of them are mutually exclusive.

Example

Let A be the event that an even number occurs, B that an odd number occurs, C that a prime number occurs.

$A = \{2, 4, 6\}$, $B = \{1, 3, 5\}$, $C = \{2, 3, 5\}$

$A \cup C = \{2, 3, 4, 5, 6\}$ the event that an even or a prime number occurs

$B \cap C = \{3, 5\}$ the event that an odd prime number occurs

$\bar{C} = \{1, 4, 6\}$ the event that a prime number does not occur

A and B are mutually exclusive.

Example

Experiment: toss a coin 3 times and observe the sequence of heads (H) and tails (T).

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

$$A = \text{event that 2 or more heads appear consecutively} = \{HHH, HHT, THH\}$$

$$B = \text{event that five heads appear} = \emptyset$$

2.1 Finite Probability Spaces

Let S be a finite nonempty sample space, say $S = \{a_1, a_2, \dots, a_n\}$. A *finite probability space* or *probability model*, is obtained by assigning to each element a_i in S a real number p_i , called the *probability* of a_i , satisfying:

- (i) each p_i is nonnegative, i.e. $p_i \geq 0$, and
- (ii) the sum of p_i 's is 1, i.e. $p_1 + p_2 + \dots + p_n = 1$.

The *probability* of an event A , written $P(A)$, is the sum of the probabilities of the elements in A . The singleton set $\{a_i\}$ is called an *elementary event*, and for notational convenience we write $P(a_i)$ for $P(\{a_i\})$.

Example

Experiment: a loaded die is tossed.

Sample space: $S = \{1, 2, 3, 4, 5, 6\}$.

The following assignment defines a probability space: $P(1) = \frac{3}{8}$, $P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{8}$.

Let A be the event that an odd number appears, and B the event that an even number appears.

$A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$

$$P(A) = P(1) + P(3) + P(5) = \frac{3}{8} + \frac{1}{8} + \frac{1}{8} = \frac{5}{8}$$

$$P(B) = P(2) + P(4) + P(6) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

Example

Experiment: let 3 coins be tossed and the number of heads observed.

Sample space: $S = \{0, 1, 2, 3\}$.

The following assignment defines a probability space: $P(0) = \frac{1}{8}$, $P(1) = \frac{3}{8}$, $P(2) = \frac{3}{8}$, $P(3) = \frac{1}{8}$.

Let A be the event that at least one head appears, and let B be the event that all heads or all tails appear.

$A = \{1, 2, 3\}$, $B = \{0, 3\}$

$$P(A) = P(1) + P(2) + P(3) = \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = \frac{7}{8}$$

$$P(B) = P(0) + P(3) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

2.2 Equiprobable Spaces

A finite probability space S where each element of S has the same probability is called an *equiprobable space*. In particular, if S contains n elements then the prob-

ability of each element of S is $\frac{1}{n}$.

For an event A of an equiprobable space S , $P(A) = \frac{\text{number of elements in } A}{\text{number of elements in } S} = \frac{|A|}{|S|}$.

Example

Let a fair die be tossed.

A = event that an odd number appears

$$P(A) = \frac{\text{number of odd numbers on a die}}{\text{number of numbers on a die}} = \frac{3}{6} = \frac{1}{2}$$

Example

Let a card be selected from an ordinary deck of 52 cards.

A = event that the card is a spade

B = event that the card is a face card (jack, queen or king)

$$P(A) = \frac{\text{number of spades}}{\text{number of cards}} = \frac{13}{52}$$

$$P(B) = \frac{\text{number of face cards}}{\text{number of cards}} = \frac{12}{52}$$

$$P(A \cap B) = \frac{\text{number of spade face cards}}{\text{number of cards}} = \frac{3}{52}$$

Example

In poker player gets 5 cards. Number of different poker hands is $C(52, 5)$.

(a) Full house: 3 cards of one rank, 2 cards of another rank.

What is the probability of receiving a full house?

1. Choose 3-card rank. (13 ways)
2. Choose 3 cards from that rank. ($C(4, 3)$ ways)
3. Choose a different 2-card rank. (12 ways)
4. Choose 2 cards from the second rank. ($C(4, 2)$ ways)

$$\Rightarrow \text{Number of different full houses is } 13 \cdot C(4, 3) \cdot 12 \cdot C(4, 2) = 3744.$$

$$\Rightarrow \text{Probability of receiving a full house} = \frac{3744}{C(52, 5)}.$$

(b) What is the probability of receiving a four of a kind?

1. Choose 4-card rank. (13 ways)
2. Choose all 4 cards from that rank. (1 way)
3. Choose 1 of the remaining cards. (48 ways)

$\Rightarrow$ Number of different four of a kinds $= 13 \cdot 1 \cdot 48 = 624$.

$\Rightarrow$ Probability of receiving a four of a kind $= \frac{624}{C(52,5)}$.

Note: four of a kind is better in poker because it occurs less frequently.

Theorem 2.1 Let A be an event. Then $P(A) + P(\bar{A}) = 1$.

Proof: Suppose that $A = \{x_1, \dots, x_k\}$ and $\bar{A} = \{x_{k+1}, \dots, x_n\}$.

$P(A) = \sum_{i=1}^k P(x_i)$ and $P(\bar{A}) = \sum_{i=k+1}^n P(x_i)$

$\Rightarrow P(A) + P(\bar{A}) = \sum_{i=1}^k P(x_i) + \sum_{i=k+1}^n P(x_i) = \sum_{i=1}^n P(x_i) = 1$ □

Note: this theorem is useful when it is easier to compute $P(\bar{A})$ than $P(A)$.

Example

Five distinct fair dice are rolled. What is the probability of getting at least one 6?

S = sample space = set of all sequences of length 5 whose elements come from the set $\{1, 2, 3, 4, 5, 6\}$.

$\Rightarrow |S| = 6^5$

A = event of getting at least one 6

$\bar{A}$ = event of getting no 6's

$|\bar{A}| = 5^5$

$P(\bar{A}) = \frac{|\bar{A}|}{|S|} = \frac{5^5}{6^5}$

So by Theorem 2.1, $P(A) = 1 - P(\bar{A}) = 1 - \frac{5^5}{6^5}$

Example

Find the probability that among n people, at least two people have birthdays on the same month and date (but not necessarily in the same year). Assume that all months and dates are equally likely, and ignore February 29 birthdays.

A = event that at least two people have the same birthday

$\bar{A}$ = event that no two people have the same birthday

$|S| = 365^n$

$|\bar{A}| = 365 \cdot 364 \cdot \dots \cdot (365 - n + 1)$

$P(\bar{A}) = \frac{365 \cdot 364 \cdot \dots \cdot (365 - n + 1)}{365^n}$

$P(A) = 1 - P(\bar{A}) = 1 - \frac{365 \cdot 364 \cdot \dots \cdot (365 - n + 1)}{365^n}$

Theorem 2.2 Let A_1 and A_2 be events. Then $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$.

Proof: Let $A_1 = \{x_1, \dots, x_i\}$, $A_2 = \{y_1, \dots, y_j\}$ and $A_1 \cap A_2 = \{z_1, \dots, z_k\}$.

In the list $x_1, \dots, x_i, y_1, \dots, y_j$ each $z_1, \dots, z_k$ occurs twice.

$$\Rightarrow P(A_1 \cup A_2) = \sum_{t=1}^i P(x_t) + \sum_{t=1}^j P(y_t) - \sum_{t=1}^k P(z_t) = P(A_1) + P(A_2) - P(A_1 \cap A_2) \quad \square$$

Example

A fair coin is tossed 3 times, and a sequence of heads and tails is observed. What is the probability that an even number of heads or at least two tails appears?

S = set of all sequences of length 3 whose elements are from the set $\{H, T\}$

So $|S| = 2^3 = 8$

A_1 = event that an even number of heads appears = $\{HHT, HTH, THH, TTT\}$

A_2 = event that at least two tails appear = $\{TTH, THT, HTT, TTT\}$

$A_1 \cap A_2$ = event that an even number of heads and at least two tails appears = $\{TTT\}$

$$P(A_1) = \frac{4}{8} = \frac{1}{2}, P(A_2) = \frac{4}{8} = \frac{1}{2}, P(A_1 \cap A_2) = \frac{1}{8}$$

$$\Rightarrow P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2) = \frac{1}{2} + \frac{1}{2} - \frac{1}{8}$$

Example

Two different colored fair dice are rolled. The sample space S consists of ordered pairs (a, b) where a and b are integers from 1 to 6. What is the probability of getting doubles (two dice showing the same number) or a sum of 6?

$$|S| = 6^2 = 36$$

A_1 = event of getting doubles = $\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$

A_2 = event of getting a sum of 6 = $\{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$

$$P(A_1) = \frac{6}{36} = \frac{1}{6}$$

$$P(A_2) = \frac{5}{36}$$

$A_1 \cap A_2$ = event of getting doubles and a sum of 6 = $\{(3, 3)\}$

$$P(A_1 \cap A_2) = \frac{1}{36}$$

$$\Rightarrow P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2) = \frac{1}{6} + \frac{5}{36} - \frac{1}{36}$$

Corollary 2.1 If A_1 and A_2 are mutually exclusive events, then $P(A_1 \cup A_2) = P(A_1) + P(A_2)$.

Proof: Assume A_1 and A_2 are mutually exclusive. Then $A_1 \cap A_2 = \emptyset$, and hence $P(A_1 \cap A_2) = 0$. So by Theorem 2.2, $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2) = P(A_1) + P(A_2)$. $\square$

2.3 Conditional Probability

Example

Suppose that we roll two different colored fair dice. What is the probability of getting a sum of 10? What if we are told at least one die is a 5?

$$|S| = 36$$

(1, 1)	(2, 1)	(3, 1)	(4, 1)	(5, 1)	(6, 1)
(1, 2)	(2, 2)	(3, 2)	(4, 2)	(5, 2)	(6, 2)
(1, 3)	(2, 3)	(3, 3)	(4, 3)	(5, 3)	(6, 3)
(1, 4)	(2, 4)	(3, 4)	(4, 4)	(5, 4)	(6, 4)
(1, 5)	(2, 5)	(3, 5)	(4, 5)	(5, 5)	(6, 5)
(1, 6)	(2, 6)	(3, 6)	(4, 6)	(5, 6)	(6, 6)

Probability of getting a sum of 10 = $\frac{3}{36} = \frac{1}{12}$ (ovals).

If we are told that at least one die is 5 (cross), the probability of getting a sum of 10 = $\frac{1}{11}$ (cross and ovals)

A probability given that some event occurred is called *conditional probability*. Suppose E is an event in a sample space S with $P(E) > 0$. The probability that event A occurs once E has occurred, the *conditional probability of A given E* , written $P(A|E)$ is defined by:

$$P(A|E) = \frac{P(A \cap E)}{P(E)}.$$

Theorem 2.3 Suppose S is an equiprobable space and A and E are events. Then $P(A|E) = \frac{|A \cap E|}{|E|}$.

$$\text{Proof: } P(A|E) = \frac{P(A \cap E)}{P(E)} = \frac{\frac{|A \cap E|}{|S|}}{\frac{|E|}{|S|}} = \frac{|A \cap E|}{|E|} \quad \square$$

Example

A fair coin is tossed 10 times, giving heads on each toss. What is the probability of a toss of heads on the eleventh toss?

Experiment: a fair coin is tossed 11 times in a row

A = event that all heads appear

E = event that on the first 10 tosses heads appear

$$\text{Want: } P(A|E) = \frac{P(A \cap E)}{P(E)} = \frac{|A \cap E|}{|E|} = \frac{1}{2}$$

Example

A pair of different colored fair dice is tossed. Find the probability that one of the dice is 2 if the sum is 6.

A = event that 2 appears on at least one die

E = event that the sum is 6

$$E = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$$

$$A \cap E = \{(2, 4), (4, 2)\}$$

$$P(A|E) = \frac{|A \cap E|}{|E|} = \frac{2}{5}$$

2.4 Independent Events

Events A and B in probability space S are *independent* if the occurrence of one of them does not influence the occurrence of the other.

In other words, B is independent of A if $P(B) = P(B|A)$.

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\Rightarrow P(A \cap B) = P(A)P(B|A) = P(A)P(B).$$

So we can use the following definition of independence:

Events A and B are *independent* if $P(A \cap B) = P(A)P(B)$; otherwise they are *dependent*.

Example

A fair coin is tossed 3 times. Let A be the event that there is at least one head, and B the event that there is at least one tail. Are events A and B independent?

$$P(A) = \frac{7}{8} = P(B)$$

$$P(A)P(B) = \frac{7}{8} \cdot \frac{7}{8} = \frac{49}{64}$$

$A \cap B$ = event that there is at least one head and at least one tail

$$P(A \cap B) = \frac{6}{8} = \frac{3}{4}$$

$P(A \cap B) \neq P(A)P(B)$ and hence events A and B are dependent.

Example

Intuitively, if we flip a fair coin twice, the outcome of the second toss does not depend on the outcome of the first toss. Let

E_H = event that head is on first toss, and

E_T = event that tail is on second toss.

Prove that E_H and E_T are independent.

$$P(E_H) = \frac{2}{4} = \frac{1}{2} = P(E_T)$$

$E_H \cap E_T$ = event that head is on first toss and tail on second

$$P(E_H \cap E_T) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(E_H) \cdot P(E_T)$$

Therefore, E_H and E_T are independent events.

Example

Joe and Alice take a final exam in math. The probability that Joe passes is 0.70 and the probability that Alice passes is 0.95. Assume that events J = “Joe passes”, and A = “Alice passes” are independent. Find the probability that Joe or Alice (or both) pass the final exam.

We want $P(J \cup A)$.

$$P(J \cup A) = P(J) + P(A) - P(J \cap A)$$

Since J and A are independent, $P(J \cap A) = P(J)P(A)$

$$\Rightarrow P(J \cup A) = P(J) + P(A) - P(J)P(A) = 0.70 + 0.95 - (0.70)(0.95) = 0.985$$

2.5 Bayes' Theorem

We can find the conditional probability that an event F occurs, given that an event E has occurred, when we know $P(E|F)$, $P(E|\bar{F})$ and $P(F)$.

Theorem 2.4 (Bayes' Theorem) Suppose that E and F are events from a sample space S such that $P(E) > 0$ and $P(F) > 0$. Then

$$P(F|E) = \frac{P(E|F)P(F)}{P(E|F)P(F) + P(E|\bar{F})P(\bar{F})}.$$

Proof: By definition of conditional probability:

$$P(F|E) = \frac{P(E \cap F)}{P(E)} \text{ and } P(E|F) = \frac{P(E \cap F)}{P(F)}.$$

$$\Rightarrow P(E \cap F) = P(F|E)P(E) \text{ and } P(E \cap F) = P(E|F)P(F)$$

$$\Rightarrow P(F|E)P(E) = P(E|F)P(F)$$

$$\Rightarrow P(F|E) = \frac{P(E|F)P(F)}{P(E)}$$

To complete the proof we need to show that $P(E) = P(E|F)P(F) + P(E|\bar{F})P(\bar{F})$.

Note that $E = E \cap S = E \cap (F \cup \bar{F}) = (E \cap F) \cup (E \cap \bar{F})$.

Since $E \cap F$ and $E \cap \bar{F}$ are disjoint, it follows that
 $P(E) = P(E \cap F) + P(E \cap \bar{F}) = P(E|F)P(F) + P(E|\bar{F})P(\bar{F})$. □

Example

Suppose that one person in 100,000 has a particular rare disease for which there is a fairly accurate diagnostic test. This test is correct 99% of the time when given to a person selected at random who has the disease; it is correct 99.5% of the time when given to a person selected at random who does not have the disease. Given this information can we find

- (a) the probability that a person who tests positive for the disease has the disease?
- (b) the probability that a person who tests negative for the disease does not have the disease?

Should a person who tests positive be very concerned that he or she has the disease?

(a)

F = event that a person selected at random has a disease

E = event that a person selected at random tests positive for the disease

We want $P(F|E)$ (the probability that a person who tests positive for the disease actually has the disease)

$$P(F) = \frac{1}{100000} = 0.00001 \text{ (since one person in 100,000 has the disease)}$$

$$P(\bar{F}) = 1 - P(F) = 1 - 0.00001 = 0.99999$$

$P(E|F) = 0.99$ (since a person who has the disease tests positive 99% of time; this is the probability of true positive)

$P(\bar{E}|F) = 1 - P(E|F) = 1 - 0.99 = 0.01$ (this is the probability of a false negative, i.e. the probability that a person who has the disease tests negative)

$P(\bar{E}|\bar{F}) = 0.995$ (since a person who does not have the disease tests negative 99.5% of time; this is the probability of true negative)

$P(E|\bar{F}) = 1 - P(\bar{E}|\bar{F}) = 1 - 0.995 = 0.005$ (this is the probability of a false positive; i.e. that a person without the disease tests positive)

So by Bayes' Theorem:

$$P(F|E) = \frac{P(E|F)P(F)}{P(E|F)P(F) + P(E|\bar{F})P(\bar{F})} = \frac{0.99 \cdot 0.00001}{0.99 \cdot 0.00001 + 0.005 \cdot 0.99999} \approx 0.002$$

(b)

$P(\bar{F}|\bar{E})$ = the probability that someone who tests negative for the disease does not

have the disease

By Bayes' Theorem:

$$P(\bar{E}|\bar{F}) = \frac{P(\bar{E}|\bar{F})P(\bar{F})}{P(\bar{E}|\bar{F})P(\bar{F}) + P(\bar{E}|F)P(F)} = \frac{0.995 \cdot 0.99999}{0.995 \cdot 0.99999 + 0.01 \cdot 0.00001} \approx 0.99999999$$

So 99.99999% of the people who test negative really do not have the disease.

By part (a), only 0.2% of people who test positive actually have the disease. Because the disease is extremely rare, the number of false positives is far greater than the number of true positives, making the percentage of people who test positive who actually have the disease extremely small. People who test positive should not be overly concerned that they actually have the disease.

3 Graph Theory

(Corresponds to selected material from Chapters 10 and 11 in the book)

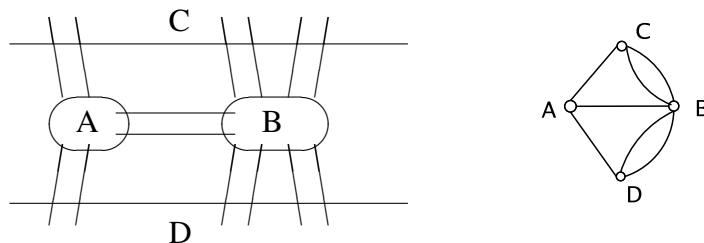
3.1 Introduction

The first paper in graph theory goes back to 1736, and some important results were obtained in the nineteenth century. Since the 1920s graph theory started developing at great speed, due to its modern applicability in diverse fields such as computer science, chemistry, operations research, electrical engineering, linguistics and economics. Here are some examples of problems that can be modeled using graphs.

Königsberg Bridge Problem

(Solved by Euler in 1736)

Two islands lying in the Pregel River in Königsberg, were connected to each other and the river banks by 7 bridges. The mayor of the town wanted to know whether it is possible for a parade to cross each of the seven bridges once and only once during the walk through the town.



Euler proved that this is not possible by proving the following more general result: a connected graph with at least one edge has an Euler cycle if and only if it has no vertices of odd degree.

Friendship Puzzle

At every party there are two people that have exactly the same number of friends at the party.

The Four Color Problem

(First appeared in London 1852)

Is it possible to color every map with 4 colors so that no two adjacent countries are colored the same?

In 1879 Kempe published a “solution” and was elected a Fellow of the Royal Society.

In 1890 Heawood published a refutation.

Kempe’s idea was that of alternating paths in the Five Color Problem and it led eventually to a proof by Appel and Haken 1976-86. Their proof makes an extensive use of a computer. No proof of this result without the use of a computer is known.

Exam Schedules

Let the vertices of a graph correspond to university courses, and put edges between those courses that have common students.

Then the notion of a chromatic number of a graph, in this example corresponds to the minimum number of time periods needed to schedule examinations without conflicts.

Marriage Theorem

If every girl in a village knows exactly k boys and every boy knows exactly k girls, then each girl can marry a boy she knows and each boy can marry a girl he knows.

Some more examples:

- How many layers does a computer chip need so that wires in the same layer do not cross?
- In what order should a travelling salesman visit cities to minimize travel time?
- What is the maximum flow per unit time from source to sink in a network of pipes?
- How can we lay cable at minimum cost to make every telephone reachable from every other?

3.2 Graph Models

A simple graph $G = (V(G), E(G))$ with p vertices and q edges consists of a *vertex set* (or a *node set*) $V(G) = \{v_1, \dots, v_p\}$ and an *edge set* $E(G) = \{e_1, \dots, e_q\}$ where each edge is an unordered pair of vertices. An edge $e = \{u, v\}$ is also denoted by uv or vu .



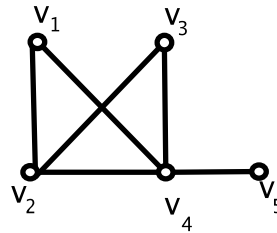
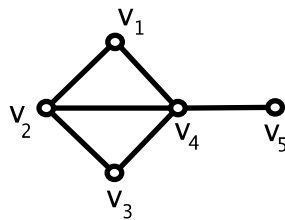
The vertices contained in an edge e are its *endpoints* (or *endnodes* or *endvertices*), and e is said to *connect* u and v . An edge $e = uv$ is said to be *incident* to its endpoints u and v . Two vertices that are endpoints of the same edge are said to be *adjacent vertices*, and two edges that are incident to the same vertex are said to be *adjacent edges*. Two adjacent vertices are also called *neighboring vertices*.

Example

$$G = (V(G), E(G))$$

$$V(G) = \{v_1, v_2, v_3, v_4, v_5\}$$

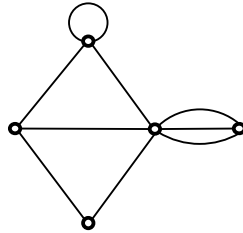
$$E(G) = \{v_1v_2, v_2v_3, v_3v_4, v_1v_4, v_2v_4, v_4v_5\}$$



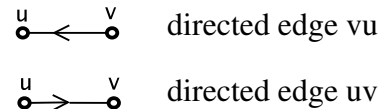
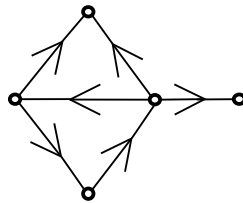
v_1 is adjacent to v_2 and v_4 , but not adjacent to v_3 and v_5
 v_2 and v_4 are neighbors of v_1

Some other graph models:

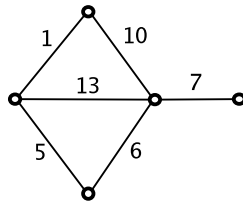
- *Multigraphs* (or *graphs* or *undirected graphs*) allow *loops* (edges joining a vertex to itself) and *parallel edges* (several edges joining the same two vertices). (Note: the terminology introduced so far for simple graphs extends to multigraphs).



- *Directed graphs* (or *digraphs*): edges are ordered pairs of vertices, and are called *directed edges* or *arcs*.



- *Weighted graphs*: each edge e is assigned weight $w(e)$



- We can also have different combinations of the above models, such as directed weighted graphs or weighted simple graphs.

Examples of modelling using graphs

1. A **communication network**: collection of cities such that certain pairs of cities can communicate with each other and others cannot.

Undirected simple graph: cities are vertices and edges represent pairs of cities that can communicate directly.

2. A **city map**.

Directed graph (since some streets are one way): vertices are points where two or more streets intersect; two intersections x and y are joined by a directed edge xy if there is a street going directly from x to y and if it is legal to travel on this street from x to y .

3. A **driving time map**: a collection of cities, some of which are joined directly by a highway.

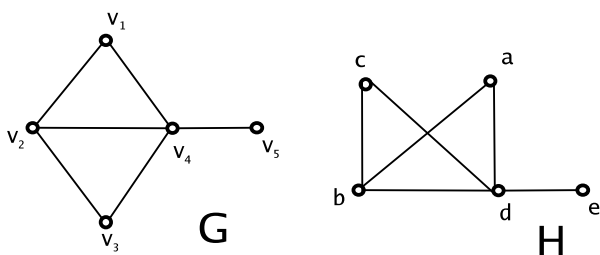
Weighted graph: vertices are cities, with one weighted edge between each two cities joined directly by a highway, where weight represents the time needed to travel that distance.

4. A **transportation network**: a network of shipping routes for transporting a commodity from its source to a large processing plant.

Weighted directed graph: source, destination and intermediate points are represented by vertices, each individual route is represented by a directed edge with an associated weight that corresponds to the maximum rate at which the commodity can be transported along this route.

There are many more examples in diverse fields.

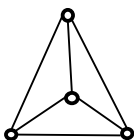
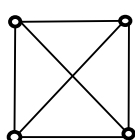
3.3 Isomorphic Graphs



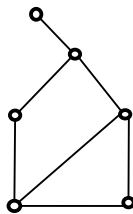
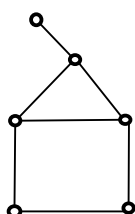
Graphs G and H look exactly the same with the exception that their vertices and edges have different labels. G and H are not identical, but isomorphic.

Let G and H be simple graphs. G is *isomorphic* to H , denoted by $G \cong H$, if there exists a bijection $f : V(G) \rightarrow V(H)$ such that $uv \in E(G)$ if and only if $f(u)f(v) \in E(H)$. Graph isomorphism for other graph models is defined similarly.

The following bijection shows that the above graphs G and H are isomorphic:
 $f(v_1) = a, f(v_2) = b, f(v_3) = c, f(v_4) = d, f(v_5) = e$.



Isomorphic graphs



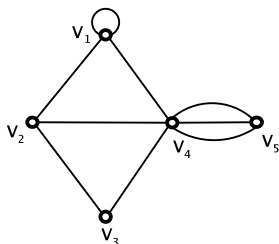
Not isomorphic (in one graph all the vertices incident to 3 edges are adjacent, and in the other they are not)

Graph isomorphism problem: determine in polynomial time whether two graphs are isomorphic. (This is a very difficult problem that is still unsolved).

Since we will mostly be interested in structural properties of graphs, we will often omit labels when drawing graphs. An unlabelled graph can be thought of as a representative of an equivalence class of isomorphic graphs. We assign labels to vertices and edges in a graph for the purpose of referring to them.

3.4 Vertex Degree

The *degree of a vertex* v in a graph G , denoted by $d_G(v)$ or just $d(v)$ (or sometimes $\deg(v)$, as in the book), is the number of edges that are incident to v (with loops counted twice, i.e. each loop on v contributes 2 to $d(v)$).



$d(v_1) = 4$
 $d(v_2) = 3$
 $d(v_3) = 2$
 $d(v_4) = 6$
 $d(v_5) = 3$

Theorem 3.1 Degree-sum Formula

If G is a graph with vertices $v_1, v_2, \dots, v_n$, then $\sum_{i=1}^n d(v_i) = 2|E(G)|$.

Proof: Each loop contributes 2 to the summation. In the summation, each edge $e = uv$ that is not a loop is counted exactly twice, once in $d(u)$ and once in $d(v)$.

□

Corollary 3.1 In any graph, the number of vertices of odd degree is even.

Proof: Let $x_1, \dots, x_n$ be vertices of a graph G with even degree and let $y_1, \dots, y_m$ be vertices of odd degree.

Let $S = d(x_1) + \dots + d(x_n)$ and $T = d(y_1) + \dots + d(y_m)$.

By Theorem 3.1, $S + T$ is even.

Since S is the sum of even numbers, S is even.

So T must be even.

Since T is the sum of m odd numbers, m must be even. □

Friendship Puzzle

At any party of two or more people, show that there are always two people with exactly the same number of friends at the party.

Proof: Let G be a simple graph with vertex set $V(G) = \{v_1, \dots, v_n\}$ denoting the people at the party.

Edges of G are defined as follows: $uv \in E(G)$ if u and v are friends.

For $i = 1, \dots, n$, $d(v_i)$ = number of friends of v_i .

Consider the following 2 cases.

Case 1: Everyone at the party has at least one friend at the party.

So for every $i = 1, \dots, n$, $1 \leq d(v_i) \leq n - 1$.

Since n variables $d(v_1), \dots, d(v_n)$ are each assigned one of $n - 1$ numbers from the set $\{1, \dots, n - 1\}$, at least two of them must have the same value.

Case 2: Someone at the party does not have any friends at the party.

For some $i \in \{1, \dots, n\}$, $d(v_i) = 0$.

So for every $i = 1, \dots, n$, $0 \leq d(v_i) \leq n - 1$.

Since n variables $d(v_1), \dots, d(v_n)$ are each assigned one of n numbers from the set $\{0, \dots, n - 1\}$, at least two of them must have the same value. □

3.5 Paths and Cycles

Caution: In literature there is a considerable variation of terminology concerning the concepts defined in this section. When consulting a book or an article, you need to make sure which definitions are used in the particular text.

Let v_0 and v_n be vertices in a graph G . A *path* in G from v_0 to v_n of length n is an alternating sequence of $n + 1$ vertices and n edges, $v_0e_1v_1e_2\ldots v_{n-1}e_nv_n$, in which edge e_i is incident on vertices v_{i-1} and v_i for every $i = 1, \ldots, n$.

Note that in a simple graph, to denote a path we just need to list the sequence of vertices since two vertices are connected by only one edge, e.g. $v_0v_1\ldots v_n$.

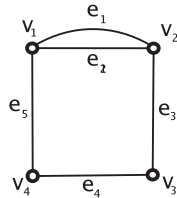
A *simple path* from vertex u to vertex v in a graph G is a path from u to v with no repeated vertices.

Lemma 3.1 *Let G be a graph, and u and v two vertices of G . There is a path from u to v in G if and only if there is a simple path from u to v in G .*

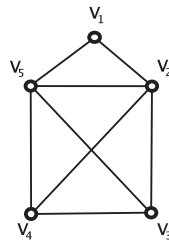
Proof: Let u and v be vertices of a graph G .

($\Leftarrow$) A simple path from u to v in G is a path from u to v , so the result trivially holds.

($\Rightarrow$) Let P be a path from u to v in G . If P has no repeated vertices, then P is a simple path and we are done. So assume that P has a repeated vertex w . Let P' be the path obtained by going along P from u to first appearance of w , and then going along P from last appearance of w to v . So P' is a path from u to v whose length is smaller than the length of P . If P' has no repeated vertices, then we are done. Otherwise, repeat this procedure. By repeating this procedure until there are no more repeated vertices, we end up with a simple path from u to v . $\square$



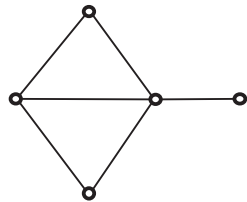
simple path of length 2: $v_1 e_1 v_2 e_3 v_3$
 simple path of length 2: $v_1 e_2 v_2 e_3 v_3$



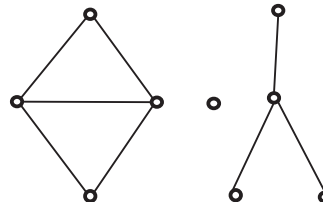
path of length 4: $v_1 v_2 v_3 v_5 v_2$
 simple path of length 3: $v_1 v_2 v_4 v_5$

Two vertices u and v of a graph G are said to be *connected* if there is a path from u to v .

A graph G is *connected* if for every pair of vertices u and v of G there is a path from u to v . Otherwise G is *disconnected*.



connected graph

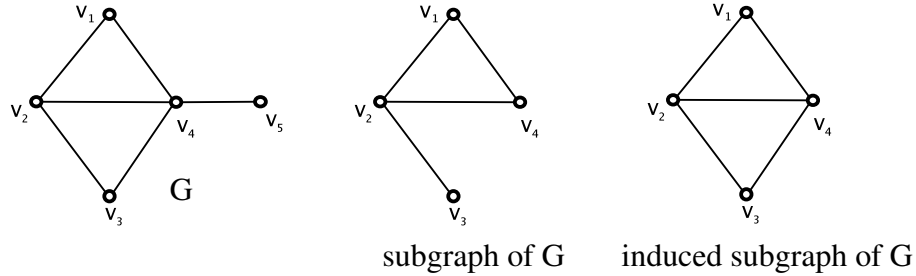


disconnected graph

A graph H is a *subgraph* of a graph G , denoted by $H \subseteq G$, if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. A subgraph H of G is a *proper subgraph* of G if $H \neq G$.

A graph H is an *induced subgraph* of a graph G , if H is a subgraph of G such that every edge of G with both endnodes in $V(H)$ is also an edge of H . We say

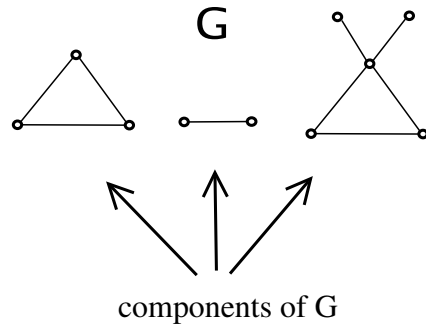
that H is the subgraph of G induced by the node set $V(H)$.



Let G be a graph. For $M \subseteq E(G)$, $G \setminus M$ denotes the graph obtained from G by removing the edges from M . So $G \setminus M = (V(G), E(G) \setminus M)$. For $S \subseteq V(G)$, $G[S]$ denotes the subgraph of G induced by S . For $S \subseteq V(G)$, $G \setminus S$ denotes the graph obtained from G by removing the vertices from S (and therefore we also remove all edges of $E(G)$ that are incident to a vertex in S). So $G \setminus S = G[V(G) \setminus S]$.

Let G be a graph. Relation “is connected to” is an equivalence relation on $V(G)$. It is reflexive, since any vertex is trivially connected to itself by a path of length 0. It is symmetric, since a path from u to v can be traversed in the opposite direction to give a path from v to u . It is transitive, since a path from u to v and a path from v to w can be combined to give a path from u to w .

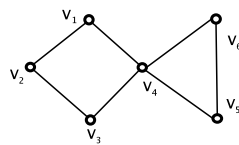
So there is a partition of $V(G)$ into nonempty subsets $V_1, \dots, V_k$ such that u and v are connected if and only if both u and v belong to the same subset V_i . For $i = 1, \dots, k$, we define G_i to be the subgraph of G induced by the node set V_i . Induced subgraphs $G_1, \dots, G_k$ are called the *components* of G . Note that connected graphs are those that have only one component.



Another way of defining components of G is as follows. A *component* (or *connected component*) of G is a maximal (in terms of the number of nodes and edges) connected subgraph of G . In other words, a connected subgraph C of G is a component of G if there does not exist a connected subgraph C' of G such that $V(C) \subseteq V(C')$, $E(C) \subseteq E(C')$ and $|V(C) \cup E(C)| < |V(C') \cup E(C')|$.

Let G be a graph and v a vertex of G . A *cycle* is a path of nonzero length from v to v with no repeated edges. The *length* of a cycle is the number of edges it contains. A *simple cycle* is a cycle from v to v in which, except for the beginning and ending vertices that are both equal to v , there are no repeated vertices.

(Note that the definition of a cycle in the book coincides with our definition of a simple cycle, and the definition of a simple circuit in the book coincides with our definition of a cycle).

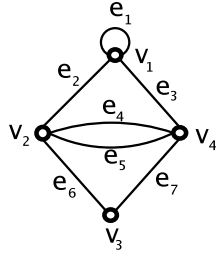


cycle: $v_1 v_2 v_3 v_4 v_5 v_6 v_4 v_1$

simple cycle: $v_1 v_2 v_3 v_4 v_1$

Euler Cycles

A cycle in a graph G that includes all of the edges and all of the vertices of G is called an *Euler cycle* (or an *Euler tour*, and in the book it is an Euler circuit). A graph is *Eulerian* if it has an Euler cycle.



Euler cycle:

$v_1 e_1 v_1 e_2 v_2 e_6 v_3 e_7 v_4 e_4 v_2 e_5 v_4 e_3 v_1$

Theorem 3.2 Euler's Theorem

A connected graph with at least one edge has an Euler cycle if and only if it has no vertices of odd degree.

Proof: Let G be a connected graph with at least one edge.

($\Rightarrow$) Suppose that G has an Euler cycle C that begins and ends at vertex v . Each time a vertex u occurs as an internal vertex of C , two of the edges incident to u are used. Since C contains every edge of G , $d(u)$ is even for every $u \neq v$. Since C starts and ends at v , $d(v)$ is even.

($\Leftarrow$) Assume that G has no vertices of odd degree.

Claim 1: For every vertex $v \in V(G)$, G contains a cycle that begins and ends at v .

Proof of Claim 1: Let $v \in V(G)$. Start building a path from v (i.e. moving from one vertex to another along the edges), without repeating edges. When the path enters a vertex $u \neq v$ through an edge, it can always leave u through an unused edge, since every vertex has an even degree and an odd number of edges incident to u have been used in reaching u . So such a walk terminates when we have reached v again, resulting in a cycle.

Let C be a cycle in G of maximum possible length.

Claim 2: C is an Euler cycle.

Proof of Claim 2: (Proof by contradiction)

Suppose not, i.e. suppose that $E(C) \neq E(G)$. Let G' be a subgraph obtained from G by removing the edges in $E(C)$. In G' every vertex has even degree. Since G is connected and $E(G') \neq \emptyset$, there is at least one edge in $E(G')$ that is incident to

some vertex u of C . By Claim 1,¹ G' contains a cycle C' that begins and ends at u . C and C' can be combined into a single cycle by breaking apart C at u to include C' . This contradicts the choice of C , and hence the claim holds. $\square$

Recognition algorithm for Eulerian graphs

Input: A graph G .

Output: YES if G is Eulerian and NO otherwise.

Step 1: Check whether G is connected (using for example Breadth First Search or Depth First Search algorithms) and contains at least one edge.

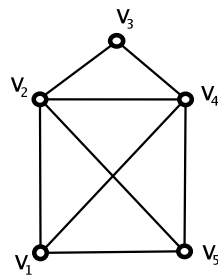
If so, then continue; otherwise return NO and stop.

Step 2: Check whether the degree of every vertex of G is even.

If it is, then return YES; otherwise return NO.

Correctness of this algorithm follows from Euler's Theorem. By following the given proof of Euler's Theorem, one can extract an algorithm that finds an Euler cycle if one exists.

In a graph G an *Euler path* is a path that includes all of the vertices of G , traverses every edge of G exactly once and it begins and ends at distinct vertices.



Euler path:

$v_1 v_2 v_3 v_4 v_2 v_5 v_4 v_1 v_5$

Corollary 3.2 *A connected graph G has an Euler path if and only if it has exactly two vertices of odd degree.*

¹Technically, we are applying Claim 1 to the connected component of G' that contains u .

Proof: Let G be a connected graph.

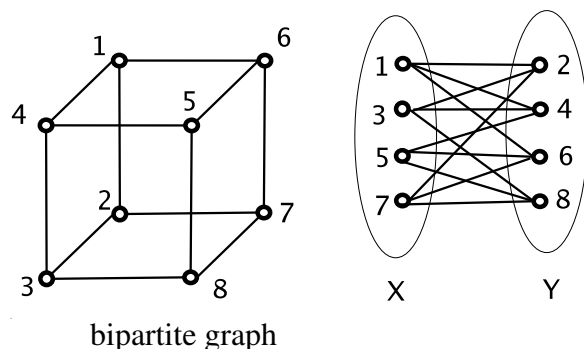
($\Rightarrow$) Suppose that G has an Euler path P that begins at u and ends at v , $u \neq v$. Each time a vertex occurs as an internal vertex of P , two of the edges incident to it are used. Since P contains every edge of G , it follows that u and v have odd degree, and remaining vertices have even degree.

($\Leftarrow$) Suppose that G has exactly two vertices u and v of odd degree. Let G' be a graph obtained from G by adding a new edge e between u and v . Note that G' is connected and has at least one edge. Also, in G' all vertices have even degree. By Theorem 3.2, G' contains an Euler cycle C . Then by removing e from C we get an Euler path. $\square$

Bipartite Graphs

A *bipartite graph* is a graph G whose vertices can be partitioned into two (possibly empty) subsets X and Y (so $X \cap Y = \emptyset$ and $X \cup Y = V(G)$) such that each edge of G has one end in X and one in Y . (X, Y) is called a *bipartition* of G .

Note that a simple graph G on one vertex, say u , is bipartite since we can let $X = \{u\}$ and $Y = \emptyset$. Another example of a bipartite graph is given in the figure.



We now characterize bipartite graphs. A *closed path* is a path that begins and ends at the same vertex. We say that a path or a cycle is *even* or *odd* depending on the parity of its length.

Note that the graph $G = (\{u, v\}, \{\{u, v\}\})$ has a closed path of positive length (e.g. u, v, u, v, u is a closed path of G of length 4), but does not have a cycle.

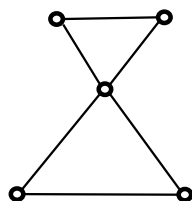
Lemma 3.2 *Every odd closed path contains an odd simple cycle.*

Proof: Suppose C is an odd closed path. We prove the result by induction on the length l of C .

Base Case: If $l = 1$, then C is a loop, which is a simple cycle of length 1.

Induction Step: Suppose $l > 1$ and the claim holds for all odd closed paths of length less than l (this is the induction hypothesis). If C has no repeated vertices (other than first = last), then C is an odd simple cycle. If vertex v is a repeated vertex in C , then we can partition C into two closed paths from v to v , say C_1 and C_2 . Since C is odd, C_1 or C_2 must be odd. Without loss of generality, C_1 is odd. Since C_1 is odd and of length less than the length of C , by induction hypothesis, C_1 contains an odd simple cycle and hence so does C . $\square$

Note that it is not true that every even closed path contains an even simple cycle.



Theorem 3.3 *A graph is bipartite if and only if it contains no odd simple cycle.*

Proof: ($\Rightarrow$) Suppose G is a bipartite graph with bipartition (X, Y) . Every path in G alternates between the nodes of X and the nodes of Y . So every return to the side of the bipartition where the path started (including to the vertex where the path started) happens after an even number of steps. Hence every simple cycle of G is even.

($\Leftarrow$) Suppose that G is a graph that has no odd simple cycle. Without loss of generality we may assume that G is connected (since if we can find a bipartition of each connected component of G , then we can merge these into a bipartition of G). Let $u \in V(G)$. By Lemma 3.2 and our assumption, for every vertex v of G , every two paths from u to v have the same parity (meaning that their lengths are

both even or they are both odd).² Partition vertices of G into X and Y as follows:
 X = set of vertices reachable from u by even paths.
 Y = set of vertices reachable from u by odd paths.

Claim: No two vertices of X are adjacent, and similarly no two vertices of Y are adjacent.

Proof of Claim: (Proof by contradiction)

Suppose that vertices v and v' from X are adjacent. Let P be a path from u to v , and Q a path from u to v' . Let P' be the path of G obtained by putting together P, Q and the edge vv' . Then P' is an odd closed path, so by Lemma 3.2, G contains an odd simple cycle, contradicting our assumption. Hence, no two vertices of X are adjacent. Similarly, no two vertices of Y are adjacent.

By Claim, (X, Y) is a bipartition of G , proving that G is bipartite. □

Note: this proof implies a simple algorithm that determines whether or not an arbitrary graph is bipartite.

Recognition algorithm for bipartite graphs

Input: A connected graph G .

Output: YES if G is bipartite (together with its bipartition), and NO otherwise.

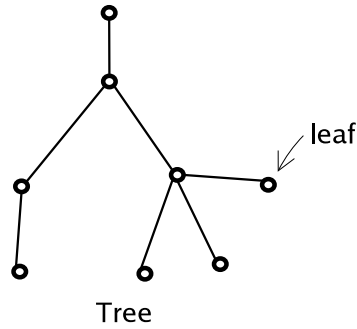
Step 1: For some vertex $u \in V(G)$, search the graph from u (using for example Breadth First Search or Depth First Search algorithms).
As vertices are discovered partition them into sets X and Y as follows:
for $v \in V(G)$, if v is reached on an even path, then put v in X , and otherwise put v in Y .

Step 2: Check whether G has an edge ab such that $\{a, b\} \subseteq X$ or $\{a, b\} \subseteq Y$.
If no such edge exists, then return YES and bipartition (X, Y) ; otherwise return NO.

3.6 Trees

A *tree* is a connected graph with no cycles. A *leaf* is a vertex of degree 1.

²Note that if two paths from u to v have different parities, together they form an odd closed path, which by Lemma 3.2 contains an odd simple cycle, contradicting our assumption.



Lemma 3.3 *Every tree with at least two vertices has at least two leaves.*

Proof: Let G be a tree with at least two vertices. Every connected graph with at least two vertices has an edge. Let P be a simple path of maximum length in G . Let u and v be the endpoints of P . Since P is of length at least 1, $u \neq v$. Suppose that u has degree greater than 1. Let u' be the neighbor of u in P , and let u'' be another neighbor of u . If u'' is in P , then there is a cycle. So u'' is not in P . Then P plus uu'' is a simple path of length longer than P , contradicting our assumption on P . Hence u is of degree 1, and similarly v is of degree 1. $\square$

Note that deleting a leaf from a tree with n vertices, results in a tree with $n - 1$ vertices.

In a connected graph G , an edge e is a *cut-edge* if the graph obtained from G by removing e is disconnected.

Lemma 3.4 *Let G be a connected graph. An edge e is not a cut-edge if and only if it belongs to a cycle.*

Proof: Let G be a connected graph, and let $e = xy$ be an edge of G .

($\Rightarrow$) Assume e is not a cut-edge. Let G' be the graph obtained from G by removing the edge e . Then G' is connected. So G' contains a path from x to y , and hence it contains a simple path P from x to y (by Lemma 3.1). But then P and e form a cycle.

($\Leftarrow$) Assume e belongs to a cycle C . Let C' be a path from x to y obtained by

removing e from C . Let G' be the graph obtained from G by removing the edge e . We now show that G' is connected. Let u and v be any two vertices of G' . Since G is connected, there is a path P from u to v in G . If P does not contain e , then P is also a path in G' . So assume P contains e . Form a path P' from u to v by deleting e from P and replacing it with C' . P' is a path in G' . So, for every $u, v \in V(G')$, there exists a path from u to v in G' . Therefore, G' is connected, and hence e is not a cut-edge. $\square$

Theorem 3.4 *For a simple graph G with n vertices, $n \geq 1$, the following are equivalent:*

- (a) G is connected and has no cycles.
- (b) G is connected and has $n - 1$ edges.
- (c) G has $n - 1$ edges and no cycles.
- (d) For every pair $u, v \in V(G)$, G has exactly one simple path from u to v .

Proof:

Proof of (a) $\Rightarrow$ (b):

Let us first clarify the statement that we want to prove. We want to prove that for every $n \geq 1$, if G is a simple graph on n vertices that is connected and has no cycles, then G is connected and has $n - 1$ edges.

In other words, we need to prove that for every $n \geq 1$, if G is a tree on n vertices, then G has $n - 1$ edges.

So the statement is of the form $(\forall n \geq 1) P(n)$, where $P(n)$ stands for ‘if G is a tree on n vertices, then G has $n - 1$ edges’.

We prove this by induction on n .

Base Case (i.e. we prove $P(1)$):

If G is a tree on $n = 1$ vertex, then since G is acyclic, it clearly has no edges (i.e. it has 0 edges).

Induction Step (i.e. we prove $(\forall n > 1) P(n - 1) \Rightarrow P(n)$):

Let $n > 1$ and assume that $P(n - 1)$ is true, i.e. assume that if G is a tree on $n - 1$ vertices, then G has $n - 2$ edges (this is the induction hypothesis). We need to

prove that $P(n)$ is true, i.e. we need to prove that if G is a tree on n vertices, then G has $n - 1$ edges. Assume G is a tree on n vertices. By Lemma 3.3, G contains a leaf v . Let G' be the graph obtained from G by removing v (and the edge incident on v). Then G' is a tree on $n - 1$ vertices. By induction hypothesis, $|E(G')| = n - 2$, and hence $|E(G)| = n - 1$. This completes the proof of (a) $\Rightarrow$ (b).

For the remainder of the proof, let G be a simple graph with $n \geq 1$ vertices.

Proof of (b) $\Rightarrow$ (c):

Assume that G is connected and has $n - 1$ edges. Delete edges from cycles of G one by one until the resulting graph G' has no cycles. Since by Lemma 3.4, no edge of a cycle is a cut-edge, G' is connected. G' is connected and has no cycles, so by (a) $\Rightarrow$ (b), G' has $n - 1$ edges. Since $|E(G)| = n - 1$, it means that no edges were removed, and hence G has no cycles. This completes the proof of (b) $\Rightarrow$ (c).

Proof of (c) $\Rightarrow$ (a):

Assume that G has $n - 1$ edges and no cycles. Suppose that G has k components $G_1, \dots, G_k$, such that for $i = 1, \dots, k$ $|V(G_i)| = n_i$. Since G has no cycles, each component satisfies (a). So by (a) $\Rightarrow$ (b), for every $i = 1, \dots, k$, G_i has $n_i - 1$ edges.

$$|E(G)| = \sum_{i=1}^k |E(G_i)| = \sum_{i=1}^k (n_i - 1) = n - k$$

Since by assumption G has $n - 1$ edges, it follows that $k = 1$, i.e. G is connected. This completes the proof of (c) $\Rightarrow$ (a).

So far we have shown that (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c) (indeed, since (a) $\Rightarrow$ (b) and (b) $\Rightarrow$ (c), by transitivity of $\Rightarrow$ it follows that (a) $\Rightarrow$ (c), and since (a) $\Rightarrow$ (c) and (c) $\Rightarrow$ (a), it follows that (a) $\Leftrightarrow$ (c); the other equivalences follow similarly). To complete the proof, we now show that (a) $\Leftrightarrow$ (d).

Proof of (a) $\Rightarrow$ (d):

Let G be a tree. Let u and v be any two vertices of G . Since G is connected, it follows that there is at least one path between any pair of vertices. So by Lemma 3.1, there is at least one simple path between u and v . (We now argue by contradiction). Suppose that there are 2 distinct simple paths from u to v , P_1 and P_2 . Since P_1 and P_2 are distinct, there is an edge $e = xy$ that is in P_1 but not in P_2 . Let G' be the graph obtained by putting together nodes and edges of P_1 and P_2 and

deleting the edge e . G' contains a path from x to y , and hence by Lemma 3.1 it contains a simple path P from x to y . But then P together with edge e is a cycle, a contradiction.

This completes the proof of (a) $\Rightarrow$ (d).

Proof of (d) $\Rightarrow$ (a):

Suppose that any 2 vertices of G are connected by a unique simple path. Then G is connected. (We now argue by contradiction). Suppose G contains a cycle C . Since G is simple, C contains at least two vertices, and hence for distinct vertices x and y of C , there exist distinct paths from x to y in C , one obtained by traversing C from x to y in one direction and the other obtained by traversing C from x to y in the other direction. Note that since C is a cycle, these two paths do not have any edges in common. Hence, by Lemma 3.1, there are distinct simple paths from x to y , a contradiction.

This completes the proof of (d) $\Rightarrow$ (a). □